

LinearSkills Academy

Class 9 — Lists

1. What is a List?

A list stores multiple values in one variable, using square brackets [].

```
fruits = ["apple", "banana", "mango"]
numbers = [10, 20, 30, 40]
mixed = ["Ali", 25, True]    # different data types allowed
```

2. Indexing & Slicing

Works just like strings — index starts from 0.

```
fruits = ["apple", "banana", "mango", "grapes"]

print(fruits[0])    # apple
print(fruits[-1])   # grapes (last item)
print(fruits[1:3])  # ['banana', 'mango']
print(fruits[::-1]) # reversed list
```

3. Common List Methods

Method	Example	Description
.append()	fruits.append("kiwi")	Adds an item at the end
.insert()	fruits.insert(1, "orange")	Adds item at a specific position
.remove()	fruits.remove("banana")	Removes item by value
.pop()	fruits.pop()	Removes and returns the last item
.sort()	numbers.sort()	Arranges list in order
.reverse()	fruits.reverse()	Reverses the list
.index()	fruits.index("mango")	Finds the index of a value
.count()	numbers.count(20)	Counts how many times a value appears
len()*	len(fruits)	Gives list length (function, not a method)

```
fruits = ["apple", "banana"]
fruits.append("mango")
print(fruits)    # ['apple', 'banana', 'mango']

fruits.remove("banana")
print(fruits)    # ['apple', 'mango']
```

4. List Comprehension

A shortcut to build a new list in one line — compresses a loop (and condition) into a single expression.

Old way:

```
squares = []
for i in range(5):
    squares.append(i * i)
print(squares)    # [0, 1, 4, 9, 16]
```

List comprehension way:

```
squares = [i * i for i in range(5)]
print(squares)    # [0, 1, 4, 9, 16]
```

With a condition:

```
even_numbers = [i for i in range(10) if i % 2 == 0]
print(even_numbers)    # [0, 2, 4, 6, 8]
```

5. Nested Lists

A list inside a list — useful for tables or grids.

```
matrix = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
]

print(matrix[0])      # [1, 2, 3]  -> first row
print(matrix[1][2])   # 6          -> row 1, column 2
```

Accessing with a loop:

```
for row in matrix:
    for item in row:
        print(item, end=" ")
# Output: 1 2 3 4 5 6 7 8 9
```

6. Mini Demo

```
students = ["Ali", "Sara", "Bilal"]
marks = [85, 92, 78]

for i in range(len(students)):
    print(students[i], "scored", marks[i])
```